

ETHAN MCCARTHY

 ethmcc.github.io

 in/ethmcc

 ethmcc

EMPLOYMENT

Lead Software Engineer

● Bill360

 2022 – 2026

 Tampa, Florida

- Architected scalable payment processing systems using C#, .NET, Node.js, Vue, and MongoDB.
- Automated cloud infrastructure and deployments via Terraform, AWS, and Azure DevOps CI/CD.
- Established testing standards across teams using automated unit and acceptance testing.

Senior Software Developer

 Two Fish Creative

 2021 – 2022

 West Palm Beach, Florida

- Led full-stack development of an enterprise medical billing platform using Python, Node.js, and Vue.
- Optimized backend performance using Redis and AWS ElastiCache for low-latency data access.
- Streamlined deployments with Docker, AWS, Pytest, and GitHub CI/CD pipelines.

Software Engineer

 Skillion Inc.

 2020 – 2021

 Bethlehem, Pennsylvania

- Led mobile and web development for an e-bike platform using Java, Android, C++, React, and TypeScript.
- Integrated AWS IoT and Google Maps Platform for real-time telemetry and route tracking.
- Implemented machine learning and vision features using OpenCV, supported by automated testing.

Visiting Assistant Professor

 University of Florida

 2018 – 2020

 Gainesville, Florida

- Published original research in theory of computation and led academic seminars.
- Built interactive teaching software and digital textbooks using React, Node.js, and MATLAB.
- Designed and delivered interactive curricula software featuring custom multimedia tools.

PERSONAL PROJECTS

I enjoy working on side-projects. You can find write-ups on my programming blog: <https://ethmcc.github.io/projects/>. These include:

- Open-source contribution to an AAC assistive device
- Progressive web application built with React for math/CS education
- Python library for efficiently solving combinatorial problems
- Native Android Application written in Java
- Hardware pomodoro timer built on the Arduino platform with C++

EDUCATION

Ph.D. Mathematics

UW–Madison

 2018

 Madison, Wisconsin

Computer science, computability and complexity, algorithmic information theory (Ph.D. 2018, M.A. 2014)

TECHNICAL SKILLS

C#  Java  Python TypeScript
Node.js  JavaScript C++  .net
 Vue  React MongoDB Redis
SQL  Jira Azure DevOps CI/CD
Terraform Docker  AWS  GCP
OOP Automated Testing Unit Testing
Integration Testing

OTHER HIGHLIGHTS

- Taught Python-based programming courses at *Art of Problem Solving* to talented high school students. (2014-2018)
- Awarded NSF Graduate Research Fellowship, 2014.

SELECTED PUBLICATIONS

- *Pointwise complexity of the derivative of a computable function*. Arch. Math. Log. (2021). DOI: f7ms
- *Cototal enumeration degrees and their applications to effective mathematics*. Proc. Am. Math. Soc. 146 (2018). DOI: f7mt

OTHER INTERESTS

-  Collegiate Sailing Instructor.
-  Snorkeling enthusiast.
-  Pickleball beginner.